

The Evolution of Post-Quantum Cryptography Science in Europe and Asia. A Comparative Trend and Cluster Analysis.

Natalie Czerwinski

Department of Information Systems and Operations Management
Vienna University of Economics and Business
Vienna, Austria
natalie.czerwinski@s.wu.ac.at

JMC Sturlese

Institute for Data, Process and Knowledge Management
Vienna University of Economics and Business
Vienna, Austria
jenny@sturlese.com

Abstract— Quantum computing poses a fundamental threat to widely deployed public-key cryptographic structures, accelerating the interest in post-quantum cryptographic (PQC) research worldwide. Existing scientometric studies have mapped the global PQC research landscape, but no dedicated regional comparison between Europe and Asia has been conducted. This contribution addresses that gap through a scientometric analysis of 3,765 publications retrieved from Scopus in the time period of 2016 to 2026. The results show that Asia has led European PQC output in cumulative publications since 2020 and has consistently widened its lead ever since. Asian research is heavily concentrated in China. European research is more evenly distributed across multiple countries. Both regions have their strongest international collaboration links with the United States, while collaborations between Europe and Asia are comparatively fewer. Thematically, lattice-based cryptography dominates both regions, but European research shows more thematically coherent research clusters, while Asia's thematic structure lacks central themes. These findings suggest that while both regions are aligned with the global direction of the NIST standardization process, they differ substantially in research concentration, collaboration patterns, and thematic development.

Keywords — *post-quantum cryptography, PQC, scientometric analysis*

I. INTRODUCTION

Sufficiently powerful, quantum computers could render widely deployed public-key cryptographic systems, such as RSA, ElGamal and ECC, insecure [1]. Post-quantum cryptography (PQC) has emerged as the leading solution to this threat and is rapidly evolving into an interdisciplinary research field that goes beyond its mathematical foundations into areas such as healthcare and policy [2-3]. In August 2024, the National Institute of Standards and Technology (NIST) officially released the first finalized PQC standards, marking a major milestone in the global effort to secure digital infrastructure against the threat of quantum computing [4].

Despite this momentum, several research gaps remain. While Hanafi and Ali [5] provide a valuable scientometric overview of the global PQC research landscape, their analysis neither offers a dedicated regional comparison between Europe and Asia nor covers publications beyond 2023, which is a significant cutoff in a rapidly evolving field.

This paper addresses both limitations through a scientometric analysis of 3,765 Scopus publications from 2016 to 2026, answering the question: *To what extent does post-quantum cryptographic research differ between Europe and Asia?* Three inquiries examine publication volume and

temporal trends, national output and international collaboration networks, and dominant research themes.

II. POST-QUANTUM CRYPTOGRAPHY

Quantum computers leverage the principles of quantum mechanics to enable speedups for specific types of mathematical problems that are computationally infeasible on classical computers [1,6]. Most critically for cryptography, Shor's algorithm can solve both the integer factorization problem and the discrete logarithm problem in polynomial time, directly threatening RSA, ElGamal, and ECC [7]. PQC addresses this threat by designing cryptographic algorithms that run on classical hardware but remain secure against both classical and quantum adversaries [8]. PQC is a landscape of approaches, each resting on a different hard problem [1]. Five main families have been developed and evaluated through the NIST standardization process.

Lattice-based cryptography is the most dominant PQC family in the NIST standardization effort. Its security rests on the hardness of problems in high-dimensional lattices, most notably the Learning With Errors (LWE) problem and its ring and module variants [9-11]. These problems become computationally infeasible as the number of dimensions grows, and no efficient quantum algorithm is known to solve them [9]. NIST finalized two lattice-based standards in August 2024: ML-KEM (based on CRYSTALS-Kyber) for key encapsulation, and ML-DSA (based on CRYSTALS-Dilithium) for digital signatures [4]. A third, FN-DSA (based on FALCON), is in draft for standardization [4].

Hash-based cryptography mainly provides digital signature schemes whose security reduces entirely to the security of a cryptographic hash function, one of the most well-understood security assumptions in the field. NIST standardized the stateless scheme SLH-DSA (based on SPHINCS+) in August 2024, alongside the stateful schemes XMSS and LMS [4].

Multivariate cryptography is also primarily a digital signature scheme family and is built on the hardness of solving systems of multivariate quadratic polynomial equations over finite fields, which is known to be NP-hard [6,8]. However, repeated classical breaks of prominent candidates, most notably the break of Rainbow in 2022 by Ward Beullens, have significantly undermined practical confidence [12-13]. UOV and the newer MAYO variant remain under evaluation in NIST's additional signature process [4,14].

Code-based cryptography is one of the oldest PQC families, introduced by Robert McEliece in 1978 [15]. It relies on the difficulty of decoding a random-looking linear error-correcting code without knowledge of its hidden structure [8,15]. Classic McEliece has withstood cryptanalytic attacks

for over four decades, though its large public key sizes have limited practical adoption [9]. In March 2025, NIST selected HQC as a backup key encapsulation mechanism alongside ML-KEM [16].

Isogeny-based cryptography relies on the hardness of computing isogenies between supersingular elliptic curves and offered extremely compact keys [8,13]. It appeared especially promising due to the absence of known quantum attacks. However, it suffered a critical setback in August 2022 when Castryck and Decru broke SIDH, the primary isogeny-based key exchange scheme, with a classical attack running in under an hour on a single CPU core [17]. SQIsign, which is not affected by this attack, remains an active candidate in NIST's additional signatures process [4].

III. METHODOLOGY

This study used scientometric analysis to map the PQC research landscape in Europe and Asia by analysing publication and citation data. The methodology is primarily based on the recommended workflow by Aria et al. [18], and draws on the bibliometric approach used by Hasija et al. [19].

Data was collected from Scopus due to its broad interdisciplinary coverage and trustable scientific sources. To extract the relevant data, the title, abstract, and author keyword fields were searched using the query: *TITLE-ABS-KEY(("PQC" OR "post-quantum cryptography" OR "post quantum cryptography") OR ("quantum-safe" OR "quantum safe")) AND PUBYEAR > 2015 AND PUBYEAR < 2027*

This search string was intentionally broad to avoid excluding relevant work that uses alternative terminology. The lower time bound of 2016 was chosen because it marks the start of the exponential publication growth in PQC, coinciding with the launch of NIST's PQC standardization competition [5]. Data was collected on April 3rd, 2026, resulting in 5,891 records prior to any cleaning or handling. However, for comparative purposes, all temporal plots in Section A are restricted to complete years up to and including 2025.

Records with missing or duplicate DOIs were removed, and publications were filtered to include only those with at least one author affiliation from a European or Asian country, based on predefined country lists. This resulted in 4,730 records remaining.

Publications were then screened first by title and then by abstract to ensure genuine relevance to PQC. Publications were excluded if they only mentioned PQC as a direction for future work, focused exclusively on classical cryptography without PQC relevance, or addressed quantum computing without a cryptographic application. Notably, the term "PQC" was a significant source of false positives, as in other fields the acronym stands for "protein quality control" or "pavement quality concrete". This two-stage screening process resulted in a final dataset of 3,765 records.

Before analysis, the final dataset was segmented into two main regional subsets (Europe and Asia) for comparative analysis. A publication was classified as European if any author affiliation was from a European country. A publication was classified as Asian if any author affiliation was from an Asian country. Publications with authors from both regions were included in both regional subsets for analysis. The resulting regional distribution of the dataset is summarized in Table I.

TABLE I. REGIONAL CLASSIFICATION OF PUBLICATIONS IN THE DATASET (N = 3,765)

Region ^a	Publications	%
Europe only	1,039	27.6
Asia only	1,934	51.4
Cross-regional (Europe and Asia)	285	7.6
Mixed (including other regions)	507	13.4

^a. Transcontinental and Central Eurasian countries (e.g., Turkey, Israel, Saudi Arabia, the South Caucasus, and Central Asian republics) were assigned to Asia. Russia and Cyprus were assigned to Europe.

All following analyses were conducted with R 4.3.3 using the bibliometrix, dplyr, and ggplot2 packages.

IV. RESULTS

The following results are based the final dataset of 3,765 publications and presented comparatively between Europe and Asia across three dimensions.

A. Publication Volume and Temporal Trends

Fig. 1 presents the annual publication volume for the combined European and Asian dataset from 2016 to 2026. It shows consistent growth with 37 publications in 2016 to a peak of 1,000 in 2025. Growth was gradual from 2016 to 2019, with annual counts staying below 200, and accelerated from 2020 onwards.

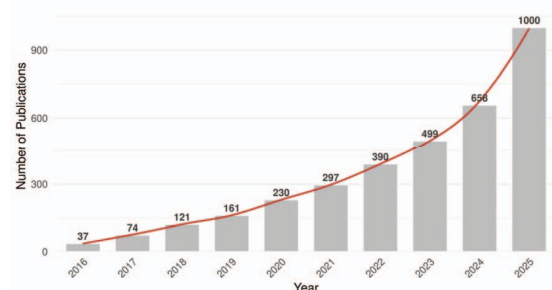


Fig. 1. Annual publication count for Europe and Asia

The annual publication count gets separated into Europe and Asia in Fig. 2. In 2016 and 2017, European output slightly outpaced Asian output, with Europe producing 66 publications compared to Asia's 56. From 2018 onwards, Asia consistently exceeds Europe each year. The gap between the two regions widened substantially from 2022, with Asia reaching 739 publications in 2025 compared to Europe's 337 in the same year.

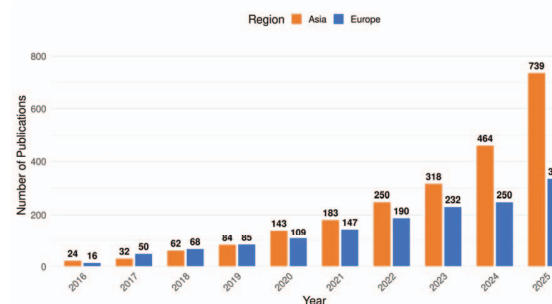


Fig. 2. Annual publication count by region

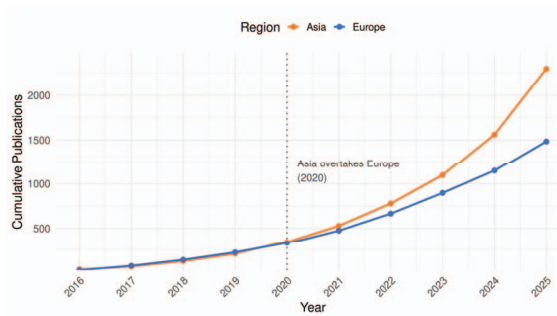


Fig. 3. Cumulative publication growth by region

Fig. 3 illustrates the cumulative publication growth for each region from 2016 to 2026. Both Europe and Asia began at near-identical cumulative counts in 2016 and remained close until 2019. As annotated in the figure, Asia overtook Europe in cumulative publications in 2020, which also marks Asia's increasing disparity to Europe's cumulative publication. By the end of the dataset, Asia had produced 2,530 publications and Europe had 1,586 publications.

B. National Output and International Collaboration

The top ten most productive countries in each region, ranked by total publication count, can be seen in Fig. 4. In Asia, China leads the ranking with 774 publications, closely followed by India with 707. A substantial gap can be identified to the next-ranked countries Japan (264) and South Korea (243). The remaining six countries each contributed fewer than 125 publications. In Europe, Germany leads with 324 publications, followed by the United Kingdom with 253 publications and France with 241 publications. Based on the data shown in the figure, it can be deduced that European output is more evenly distributed across the top ten countries, with the gap between the leading and tenth-ranked country (Sweden with 59) being proportionally smaller than in Asia (Malaysia with 57). Notably, the top three Asian countries together account for more publications than the entire European top ten combined.

The international collaboration network can be seen in Fig. 5. It illustrates the co-authorship links between countries across three collaboration types: Asia–Other (orange), Europe–Asia (green), and Europe–Other (blue). Europe–Asia collaborations are concentrated between Western Europe and East Asia, with most green lines connecting to China, India

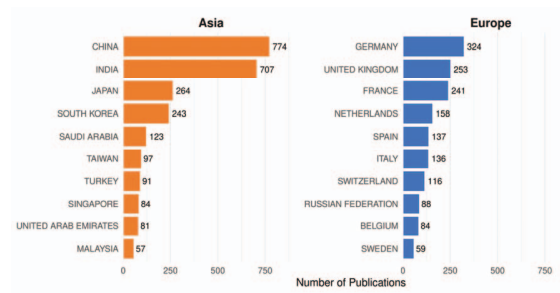


Fig. 4. Top 10 most productive countries by region

and Taiwan. The green lines also appear to originate primarily from a small number of Western European countries, with notable lines originating from the United Kingdom. From the figure, it is also visible that a significant amount of Europe–Asia collaborations lead to the Middle East, especially Israel and Saudi Arabia. Europe and Asia show their densest external collaboration ties with the United States, making it the most connected external collaborator for both regions. Moreover, the map shows notable lines connecting to Oceania, particularly Australia and significant links between Europe and New Zealand. Only a few collaborations are visible between either of the two regions and African or South American countries.

C. Dominant Research Themes

A thematic map shows the most relevant issues in a research field according to the author's keywords and helps to understand the current state and future direction of research development [20]. Fig. 6 and 7 illustrate the thematic maps for Asia and Europe respectively. Each map plots keyword clusters on two axes: Centrality along the x-axis measures how central a theme is to the entire research field, and density along the y-axis measures how well-developed it is. This produces four quadrants: motor themes, niche themes, basic themes, and emerging themes. [20] Themes in the upper-right quadrant (motor themes) are both highly developed and central to the research field [20–21]. Clusters in the upper-left quadrant are the niche themes and are well-developed but are highly specialized, making them not central to the broader field [21]. Basic themes are in the lower-right quadrant and indicate high centrality but low density [21]. These themes are fundamental topics that are essential to the field but are not yet fully developed [21]. Themes in the lower-left quadrant

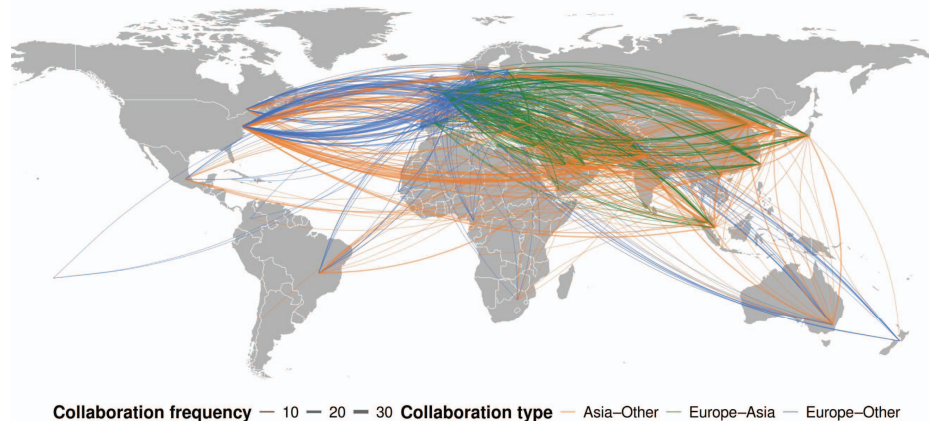


Fig. 5. International collaboration network

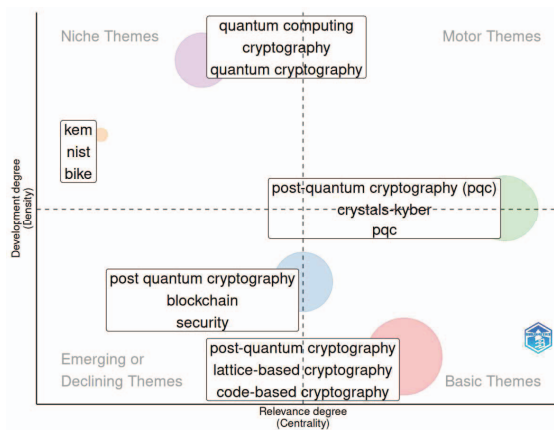


Fig. 6. Thematic map of Asia

(emerging or declining themes) are neither well-developed nor central to the research field and are either just starting to gain traction or fading in relevance [20-21].

In the Asian thematic map (Fig. 6), no cluster is clearly positioned in the motor themes quadrant. Two clusters emerge in the niche themes quadrant: “quantum computing, cryptography, quantum cryptography” and “KEM, NIST, BIKE”. The cluster “post-quantum cryptography (PQC), CRYSTALS-Kyber, PQC” sits on the boundary between the motor and basic themes quadrants. The cluster “post-quantum cryptography, lattice-based cryptography, code-based cryptography” is positioned in the basic themes quadrant, and the cluster “post quantum cryptography, blockchain, security” appears on the boundary between basic themes and emerging or declining themes quadrant.

In the European thematic map (Fig. 7), two clusters appear as motor themes: “post-quantum cryptography, lattice-based cryptography, digital signature” and “security, cryptography, post-quantum cryptography (PQC)”. Both are highly developed and central to European PQC research. The cluster “hash-based signatures, SPHINCS+, XMSS” is positioned in the niche themes quadrant. In the basic themes quadrant, the cluster “FPGA, PQC, CRYSTALS-Kyber” can be identified. Two clusters appear in the emerging or declining themes quadrant: “code-based cryptography, cryptanalysis, side-channel attacks” and “side-channel attack, public-key cryptography, public key cryptography”.

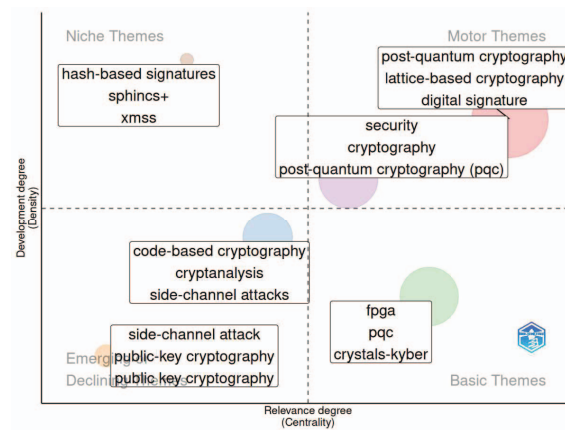


Fig. 7. Thematic map of Europe

sidechannel attacks” and “side-channel attack, public-key cryptography, public key cryptography”.

A key difference between Europe and Asia is that Europe has well-positioned motor theme clusters, while Asia does not have any clusters clearly positioned in that quadrant. Notably, in Fig. 6 it can be seen that “lattice-based cryptography” is a far less developed theme in Asia than in Europe. Additionally, Europe’s niche themes are algorithm-specific and narrow, while Asia has two niche clusters, one of which is broader and less-algorithm specific and another that is standardization-oriented.

Fig. 8 presents the publication counts for each PQC algorithm family over time, split by Europe and Asia. Lattice-based cryptography is the dominant approach in both regions across the entire observation period. In Asia, lattice-based publications grow from single digits in 2016 to approximately 250 in 2025. A steep acceleration can be identified from 2020 onwards, much more than all other approaches. In Europe, lattice-based publications grow more moderately over time, reaching a peak of approximately 75 publications in 2023 before declining. The other approaches remain comparatively low in both regions, with a slight upward trend in Asia in the year 2025. While code-based cryptography had its peak in Europe in 2023 with approximately 30 publications, it has since shown a noticeable decline. In both regions, the

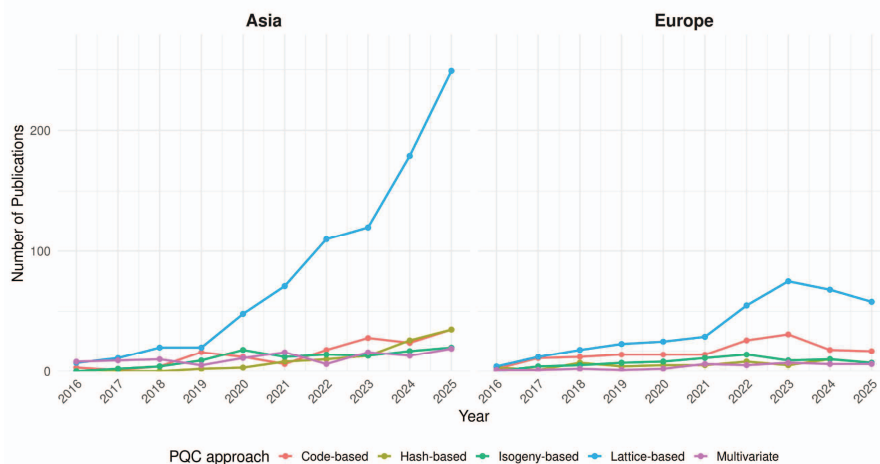


Fig. 8. PQC approaches over time

dominance of lattice-based cryptography is the most significant, though the scale of that dominance is far greater in Asia than in Europe.

V. DISCUSSION

The results presented reveal several notable differences between European and Asian PQC research across all three dimensions.

The widening gap between Asian and European publication counts from 2022 onwards, visible in Fig. 2, coincides with two significant developments in the NIST standardization competition. The third round of the NIST PQC standardization process began in July 2020, with seven finalists and eight alternate candidates selected for evaluation, and the announcement of algorithms advancing to the fourth round in July 2022 [22]. The announcement of the algorithms selected for standardization in 2022 may have prompted implementation and analysis research. Alternatively, the widening gap may be a consequence of an increase in research funding and attention for PQC across Asian countries in this period.

The international collaboration map in Fig. 5 shows that both Europe and Asia collaborate most with the United States. This underlines the centrality of the NIST standardization process.

The dominance of lattice-based cryptography in the PQC approaches overtime chart (Fig. 8) is consistent with its outcome in the NIST standardization process. ML-KEM and ML-DSA are the primary NIST standards [4], and the research community's focus on lattice-based approaches reflects this. However, the scale of lattice-based research in Asia is far greater than in Europe. This may be because of different research funding. This raises another interesting research question for potential future investigations, as funding data was not examined in this contribution.

VI. CONCLUSION

Our paper focuses on examining the extent to which post-quantum cryptographic research differs between Europe and Asia through a bibliometric analysis. Both regions show strong growth in PQC research output, though Asia has led in annual and cumulative publication counts since overtaking Europe in 2020, with the gap widening consistently. Asian output is heavily concentrated both geographically, with China and India dominating. European output is more evenly distributed across countries and institutions, with Germany, the United Kingdom, and France leading. Both regions collaborate most with the United States, while direct collaboration between European and Asian authors remains comparatively smaller. Thematically, lattice-based cryptography dominates both regions but far more dramatically in Asia. European PQC research illustrates highly developed motor theme clusters centered on lattice-based cryptography and digital signatures. Asia's motor themes remain empty, indicating that the thematic landscape is more scattered. To conclude, Europe and Asia differ most distinctly in terms of output volume, geographical concentration, and thematic maturity, while sharing a common research focus on lattice-based cryptography.

A. Limitations

Several limitations should be acknowledged. Publications with authors from both Europe and Asia were included in both regional subsets for analysis. A further limitation, author

keywords are recorded in Scopus as submitted by the author and therefore, different spellings of the same concept are treated as distinct terms by the thematic mapping algorithm. This is why "post-quantum cryptography", "post-quantum cryptography (PQC)" and "PQC" appear as separate keywords and not the same concept.

B. Further Research Perspectives

Future research could examine PQC funding allocations in Europe and Asia to explain the observed volume and thematic differences. A dedicated study of China's PQC research landscape would also provide deeper insight into the structural reasons behind its dominance at every level of analysis.

VII. APPENDIX

All R scripts for this study are publicly available in a GitHub repository. The repository can be accessed at the following URL: <https://github.com/nacz77/pqc-bibliometric-analysis.git>

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